University of Central Punjab

Faculty of Information Technology and Computer Science

Course Title: Software Construction & Development
Course Code: SESD3242

Assignment 1

Name	ID	**Contribution in the assignment** (Specifically mention the part which you have done)
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CLO	**At the end of the course the students will be able to:**	**BT Level***	**PLO**
CLO1	Understand key concepts and activities for software construction and development	C2	2

Deadline: 30-March-26 **Total Marks= 50**

Instructions:

1. **Late submission will cause 20% deduction. No submission after 2 Days.**
2. This assignment is a group assignment. 2 members in each group. Individual submissions will not be accepted.
3. **ASSIGNMENT SHOULD BE HANDWRITTEN. (No pages from register please.)**
4. **Attach this title page with your submission. (10 marks)**
5. **Upload it on the portal on 30-March-26_at 11:55pm**
6. **Plagiarism is seriously your own choice.**

Assignment 1

(Question 1)

(Part A)

How developer use psp practices

When a developer is working on the module of the project management system using PSP they go through these phases.

Planning Phase :

In begining the developer carefully understands what the module is supposed to do for example if the module is about task tracking they think about all features like adding tasks, updating status etc Then they estimate how much time it will take and break the work into small steps They also look at their past work to make better time estimates.

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Developement phase:

After planning the developer starts coding step by step while working they keep track of how much time they are spending on each task. They also note down any mistakes they make during coding like logic errors or syntax issues.

Review Phase:

Before testing the developer review their own code They check for common mistakes improve logic and make sure everything is correct. This helps in catching errors early instead of waiting for testing After that testing is done to ensure the module works properly.

Part (B)

Challenges during development

During development a developer can face many problems one common challenge is poor time estimation sometime task take longer time than expected PSP solves this by using time tracking so the developer learns from previous tasks and improve features estimates.

Another issues is frequent bugs or defects if errors are not tracked the same mistakes keep happening PSP help here by maintaining a defect log where every mistake is recorded This helps the developer understand

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patterns and avoid repeating them.

Also, developers sometimes feel overloaded or confused when tasks are not clearly structured. PSP planning breaks work into smaller parts, making it easier to manage.

(Part C)

How PSP improves quality, productivity and reliability

PSP improves software quality because developers review their code carefully and remove defects early. This means fewer errors reach the final product.

It increases productivity because developers learn how to manage their time better instead of wasting time fixing big problems later they handle small issues early.

It also improves project reliability since work is properly planned and tracked; deadlines are more realistic and results are more predictable. The team can trust that modules will work correctly and be delivered on time.

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(Question 2)

How scrum is applied

As a project manager I would organize the team using scrum in a structured way.
First, all project requirement are listed in a product backlog. These are prioritized based on importance.
Then, the team works in short time periods called sprint
In each sprint, we select some tasks from the backlog and complete them.
Everyday the team has a short meeting called a daily stand up where each member shares what they did what they will do and if they have any problems.

At the end of the sprint, we do a sprint review to show what has been completed and a retrospective to discuss what went well and what can be improved
This process continues until the whole system is completed.

(Part B)

Development challenges

One big challenge is changing requirement. For example the university may want new features later scrum handles this easily because changes can be added to

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the backlog and done in future sprints
Another issue is lack of communication in the team scrum solve this through daily meeting while everyone stays updated.

sometimes teams face delays, In scrum there issues are quickly discussed in daily standups so they can be solved early.
Also large projects can feel overwhelming but scrum divides them into small sprints making the work more manageable.

(Part C)

Scrum roles, events and artifacts.

The **Roles** are
product owner : decides what features are more important
Scrum master : helps the team follow scrum and removes obstacles
Developement Team : build the product.

There roles clearly defines responsibilities, so there is less confusion.

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Scrum events like sprint planning, daily meeting, ... improves collaboration because everyone communicates regularly.

Scrum artifacts like product backlog and sprint backlog provide transparency, meaning everyone can see what work is pending and what is completed.



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